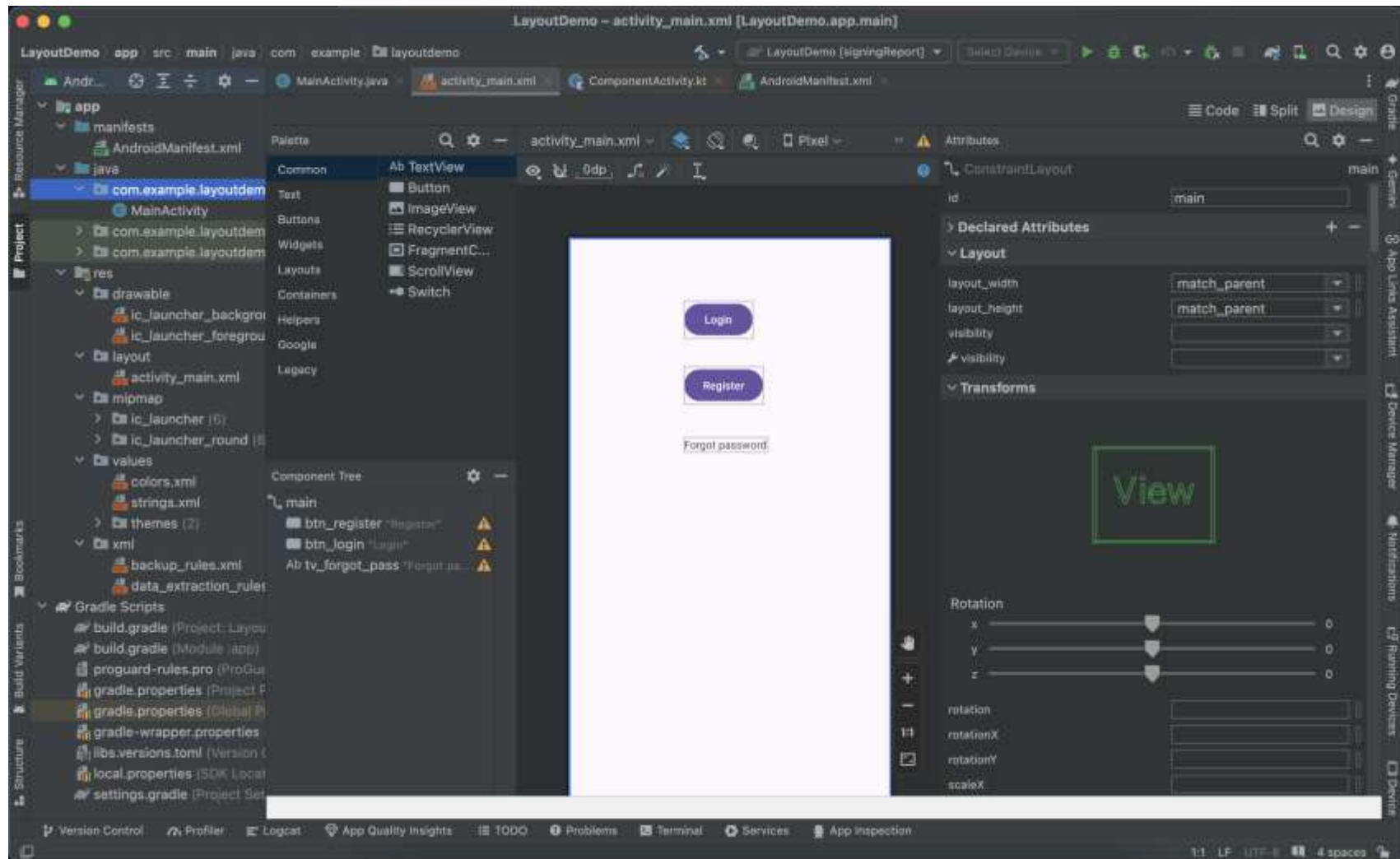


Android Application Structure

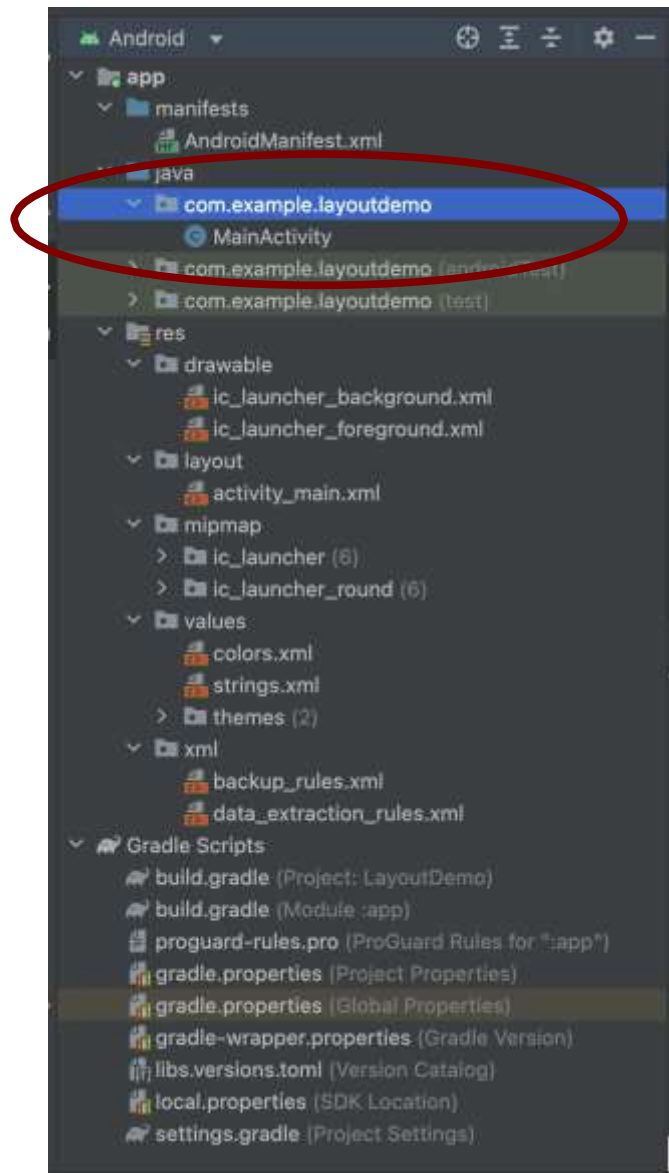
Contents

- Android project structure
- Build the first application
- Android application components

Android project structure



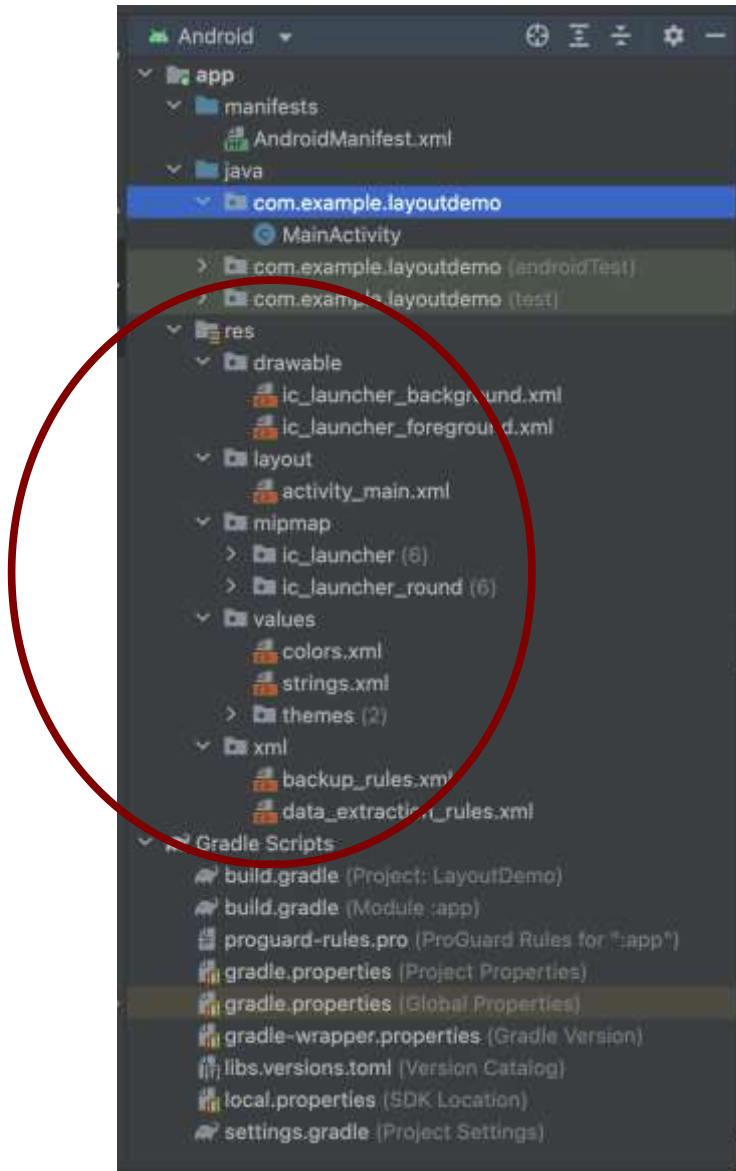
Android project structure



Contains your stub Activity file, which is stored at `src/your/package/ActivityName.java`.

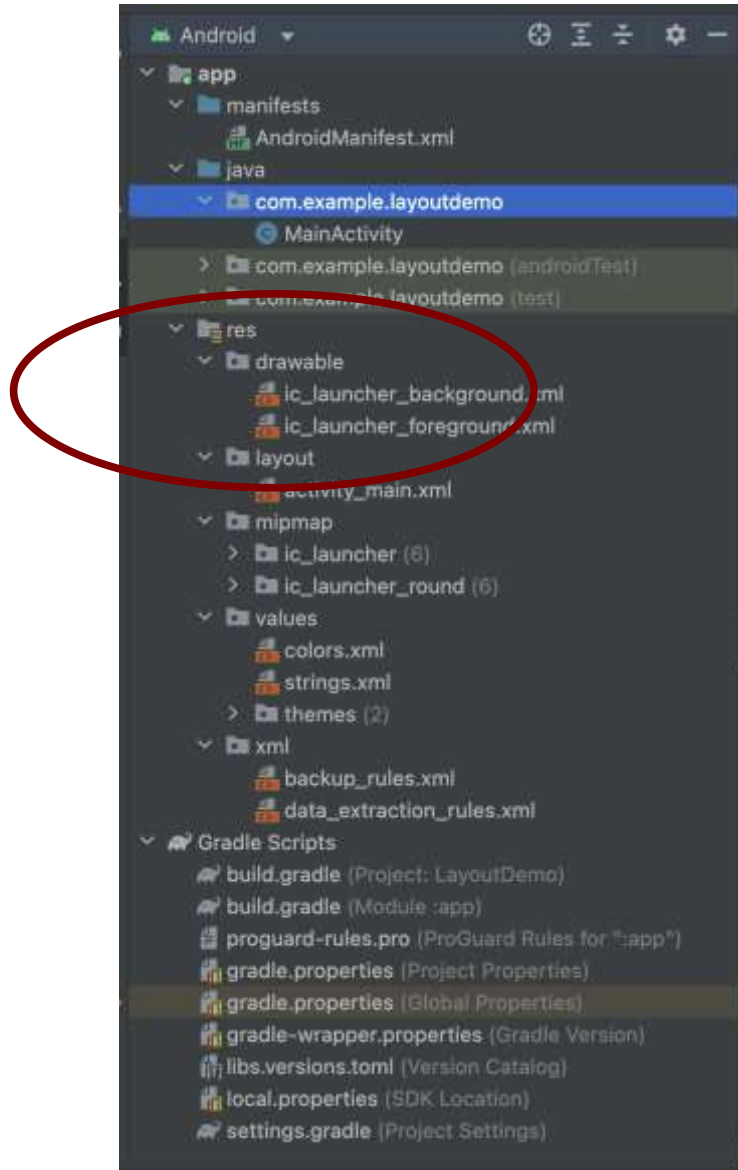
All other source code files go here as well.

Android project structure



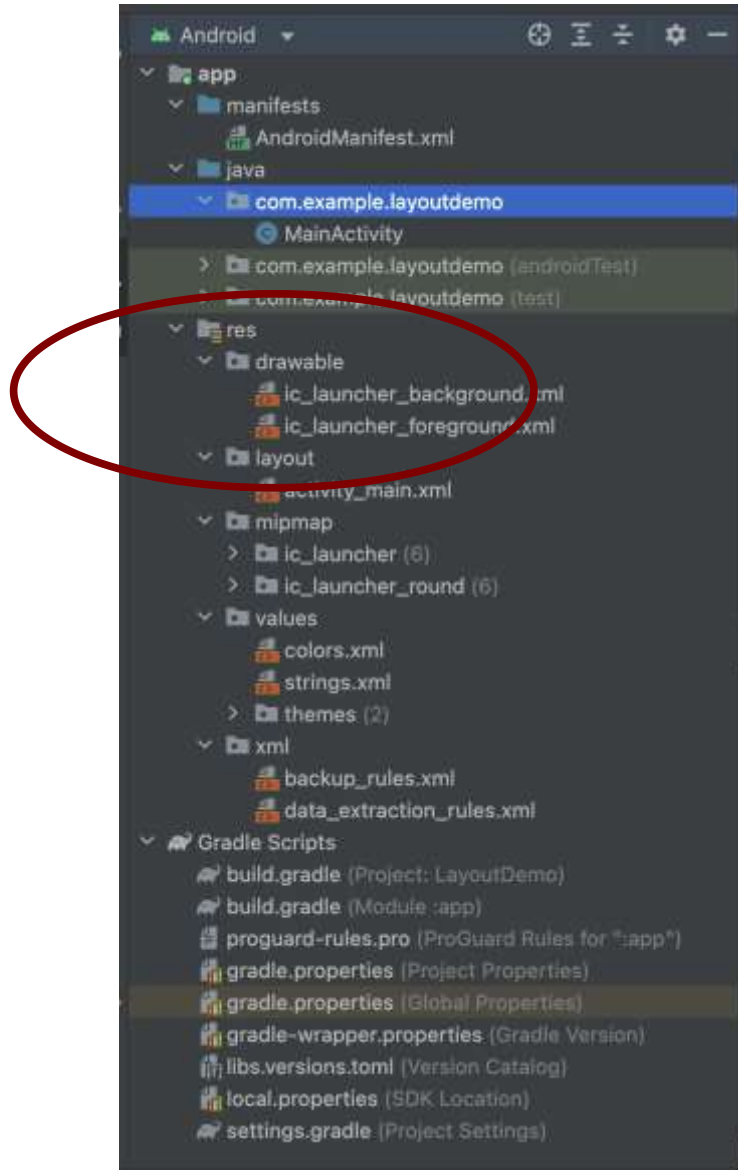
Contains application resources, such as drawable files, layout files, and string values, animation,

Android project structure



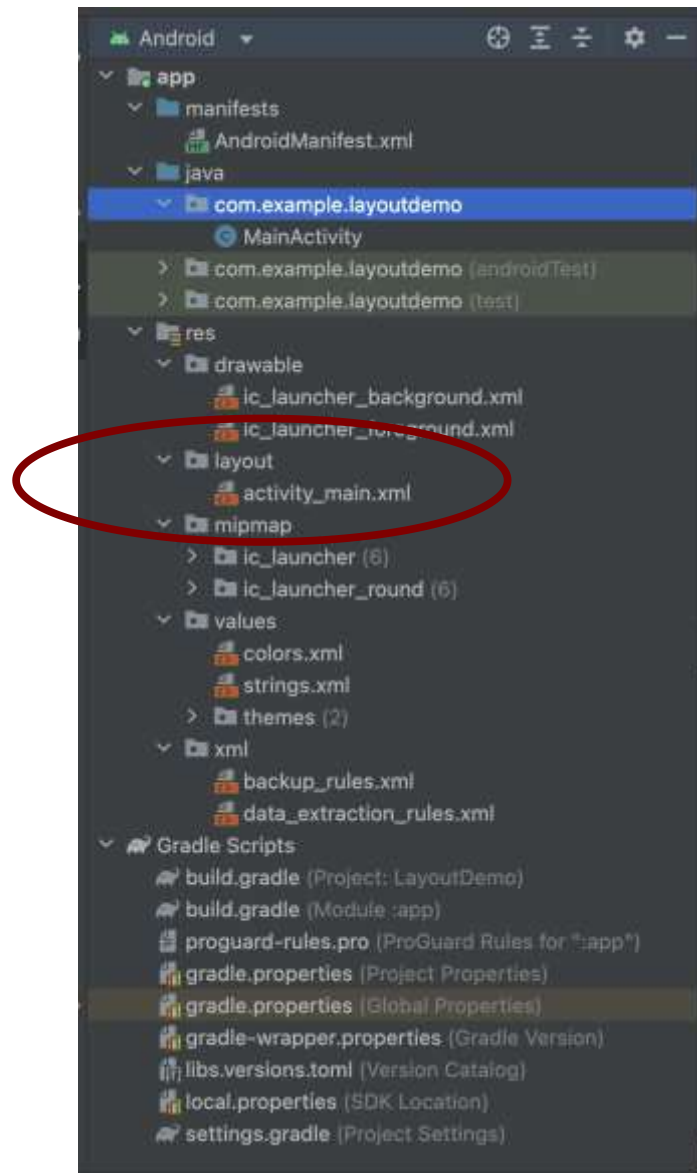
For bitmap files (PNG, JPEG, or GIF), 9-Patch image files, and XML files that describe Drawable shapes or a Drawable objects that contain multiple states (normal, pressed, or focused)

Android project structure



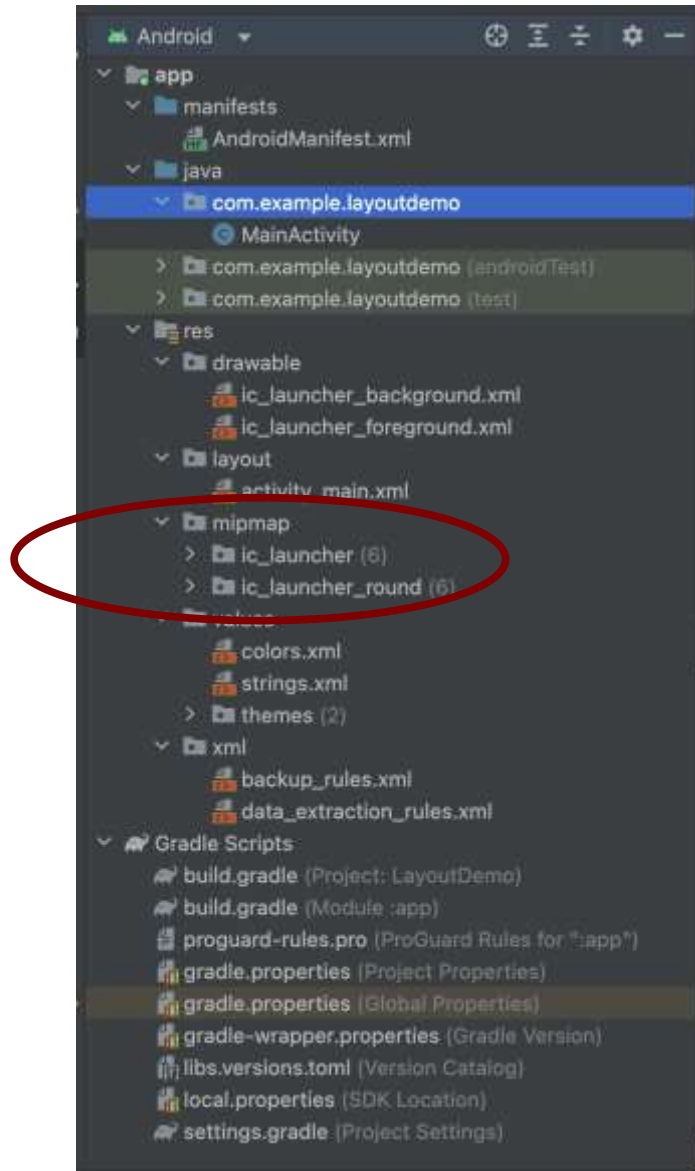
For bitmap files (PNG, JPEG, or GIF), 9-Patch image files, and XML files that describe Drawable shapes or a Drawable objects that contain multiple states (normal, pressed, or focused)

Android project structure



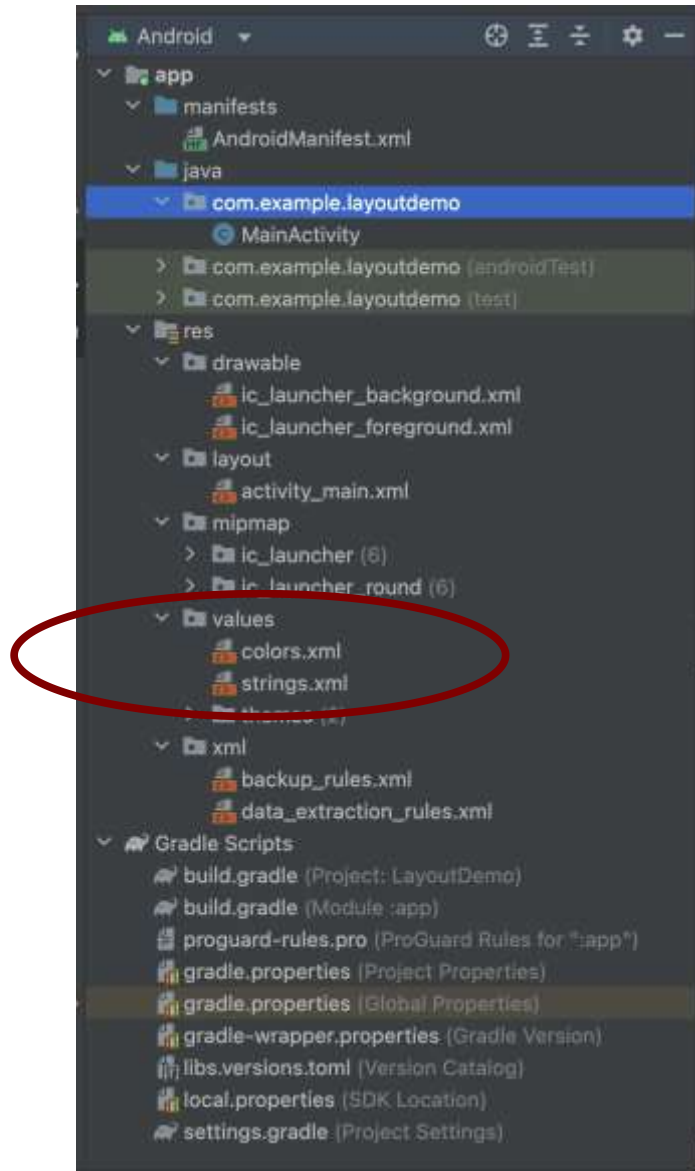
XML files that are compiled into screen layouts (or part of a screen)

Android project structure



The mipmap folder allows you to include different versions of your app's launcher icon, optimized for each screen density. This ensures that the icon appears sharp and clear on all devices, regardless of screen resolution.

Android project structure

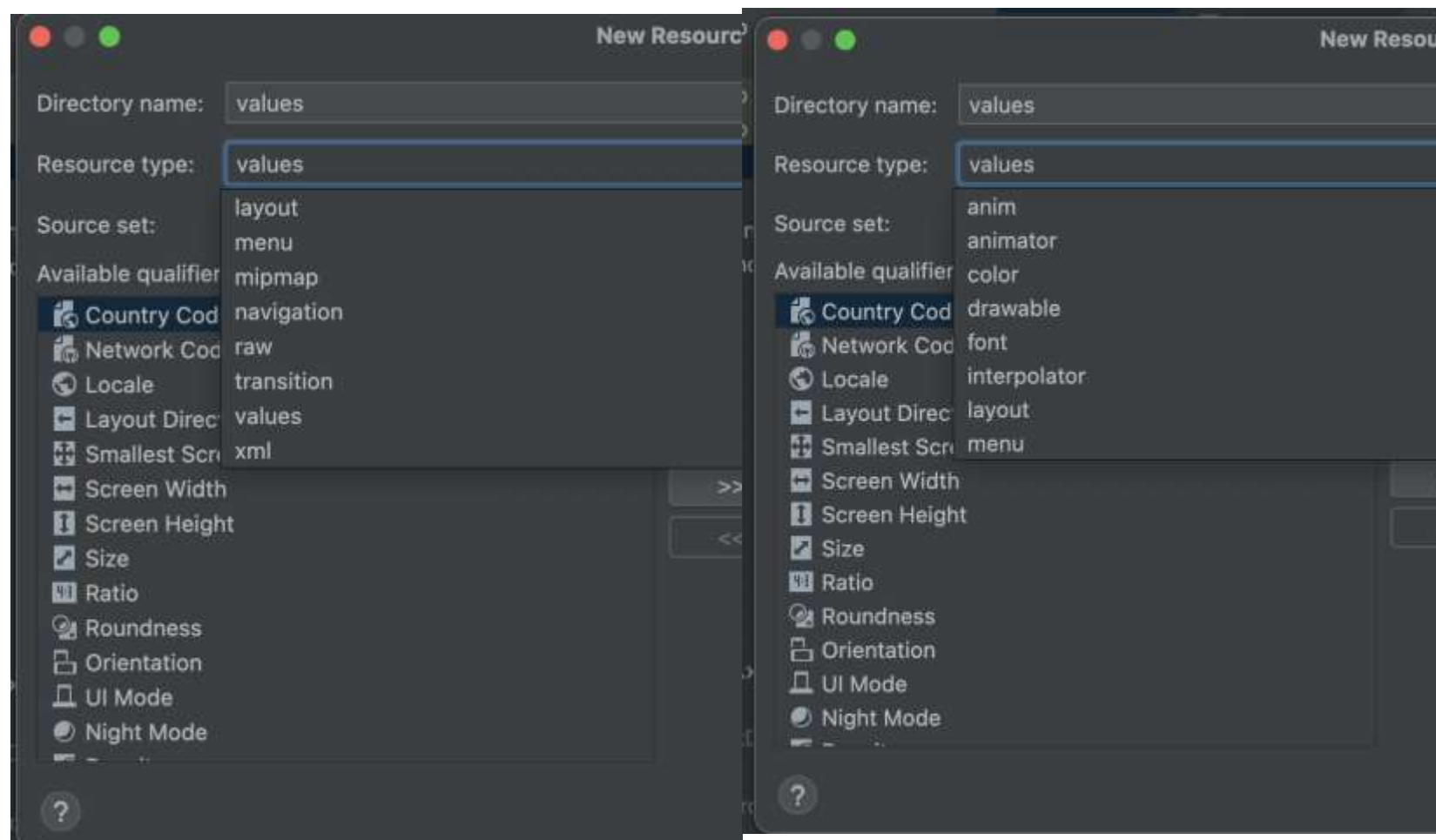


For XML files that are compiled into many kinds of resource

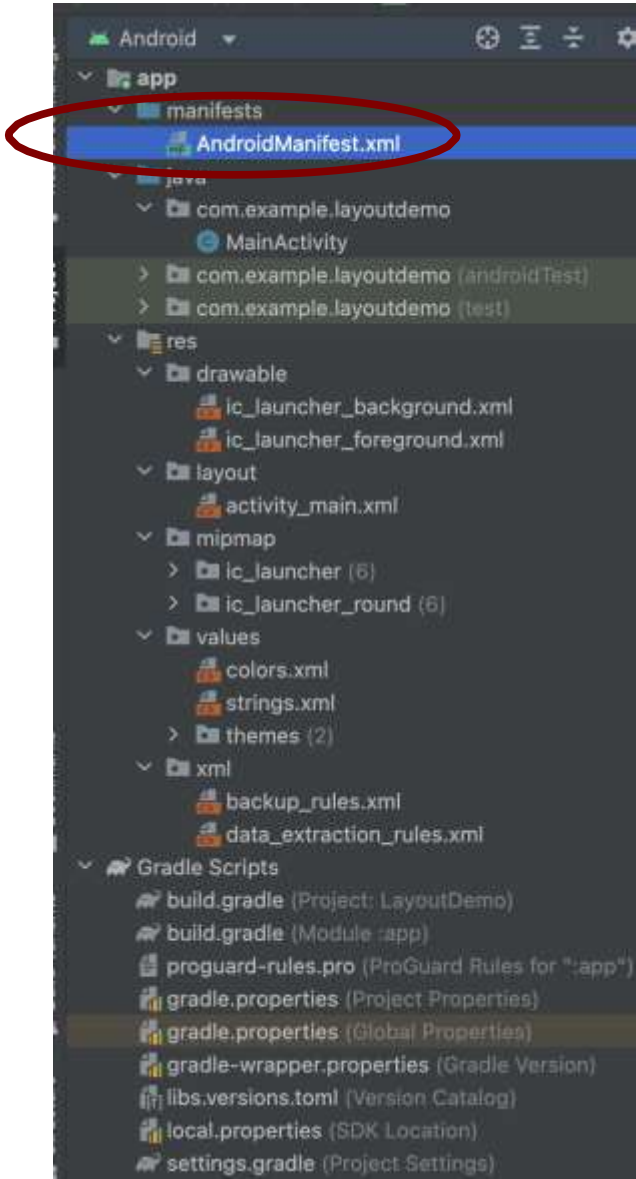
Android project structure

Create other sub resource folders

- Asset folder: For arbitrary raw asset files. Saving asset files here instead of in the assets/ directory only differs in the way that you access them.
- Menu folder: For XML files that define application menus
- Etc.

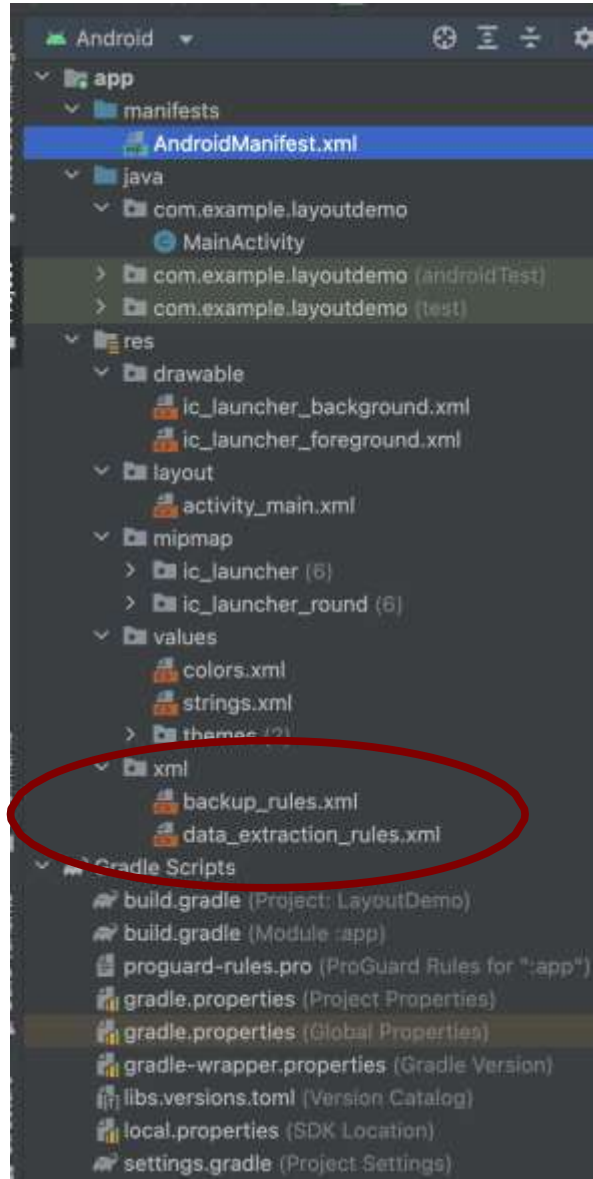


Android project structure



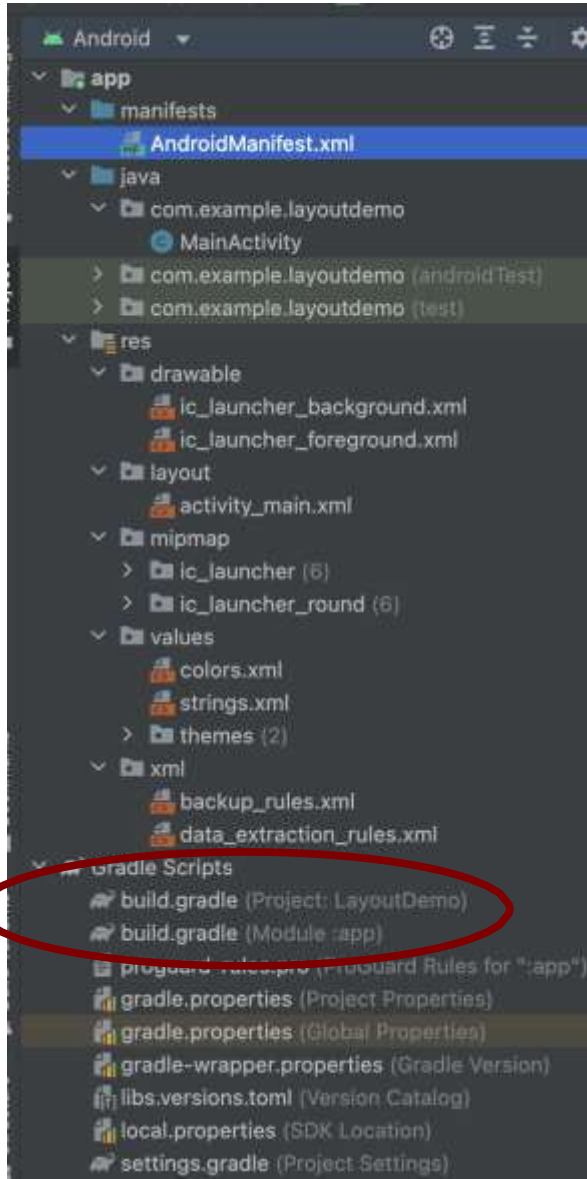
The control file that describes the nature of the application and each of its components.

Android project structure



For miscellaneous XML files that configure application components.

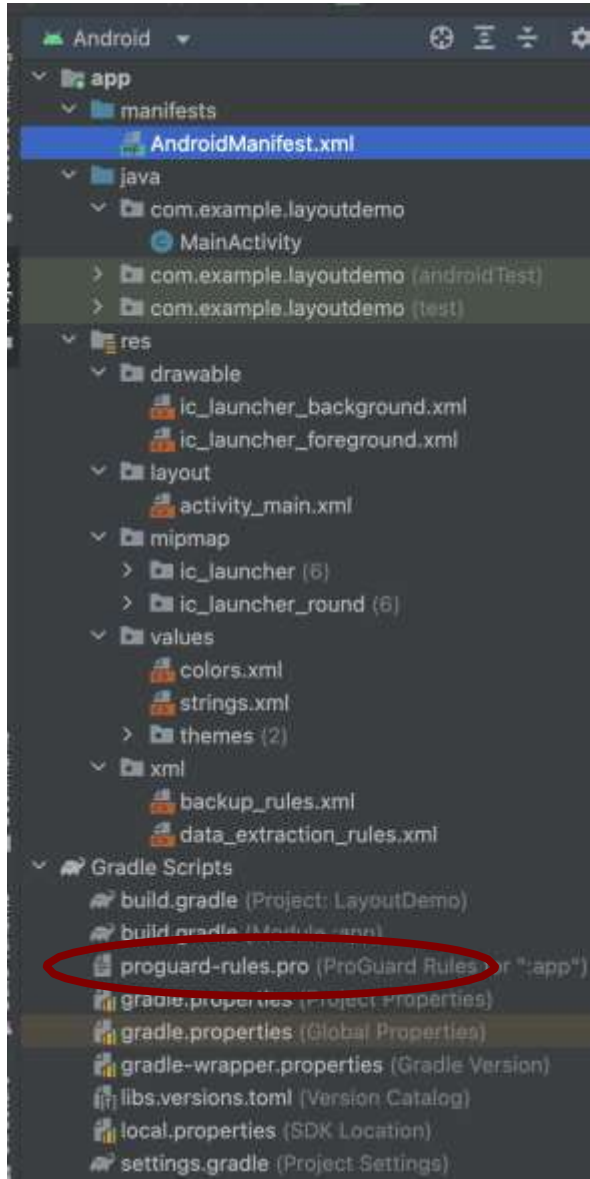
Android project structure



Two build.gradle files

- Level project: config for all project
- Level module: config for modules in project. Each module has one build.gradle file

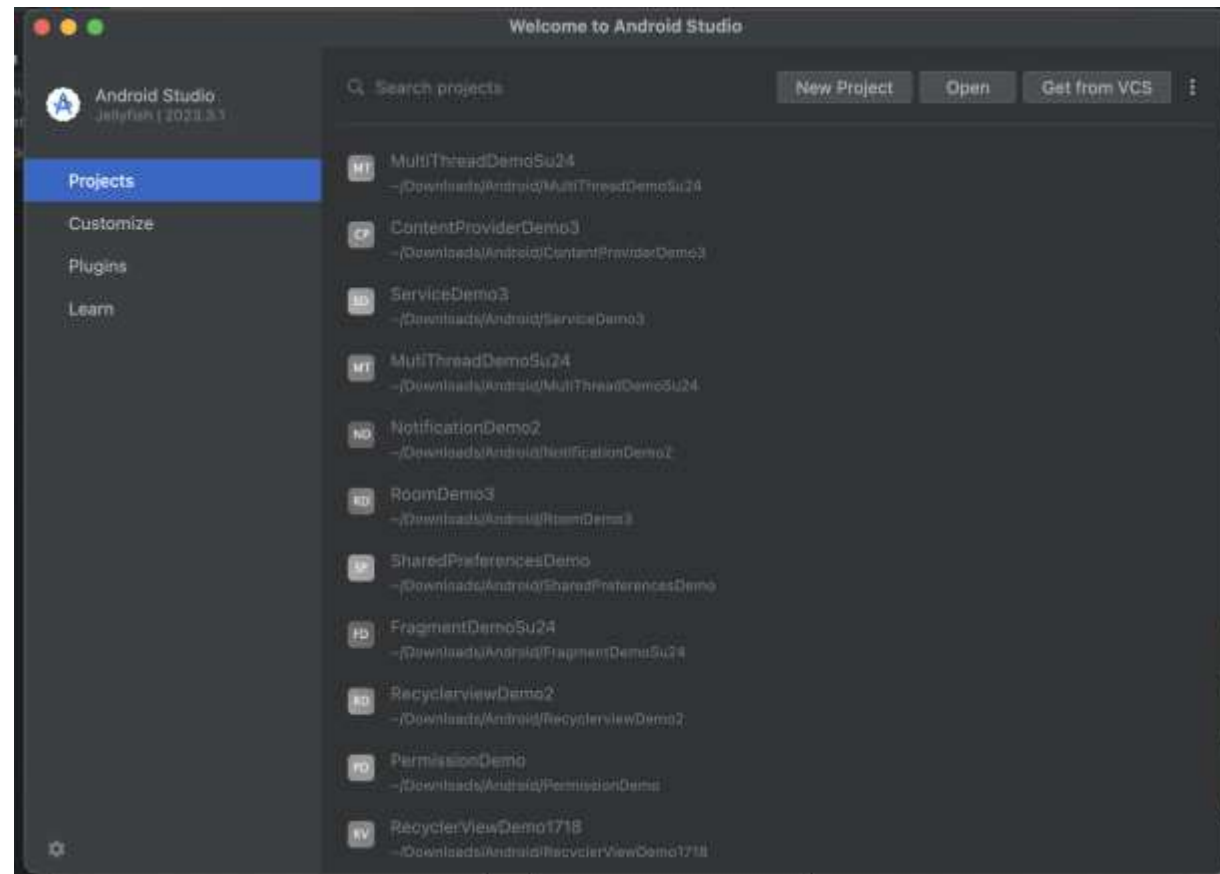
Android project structure



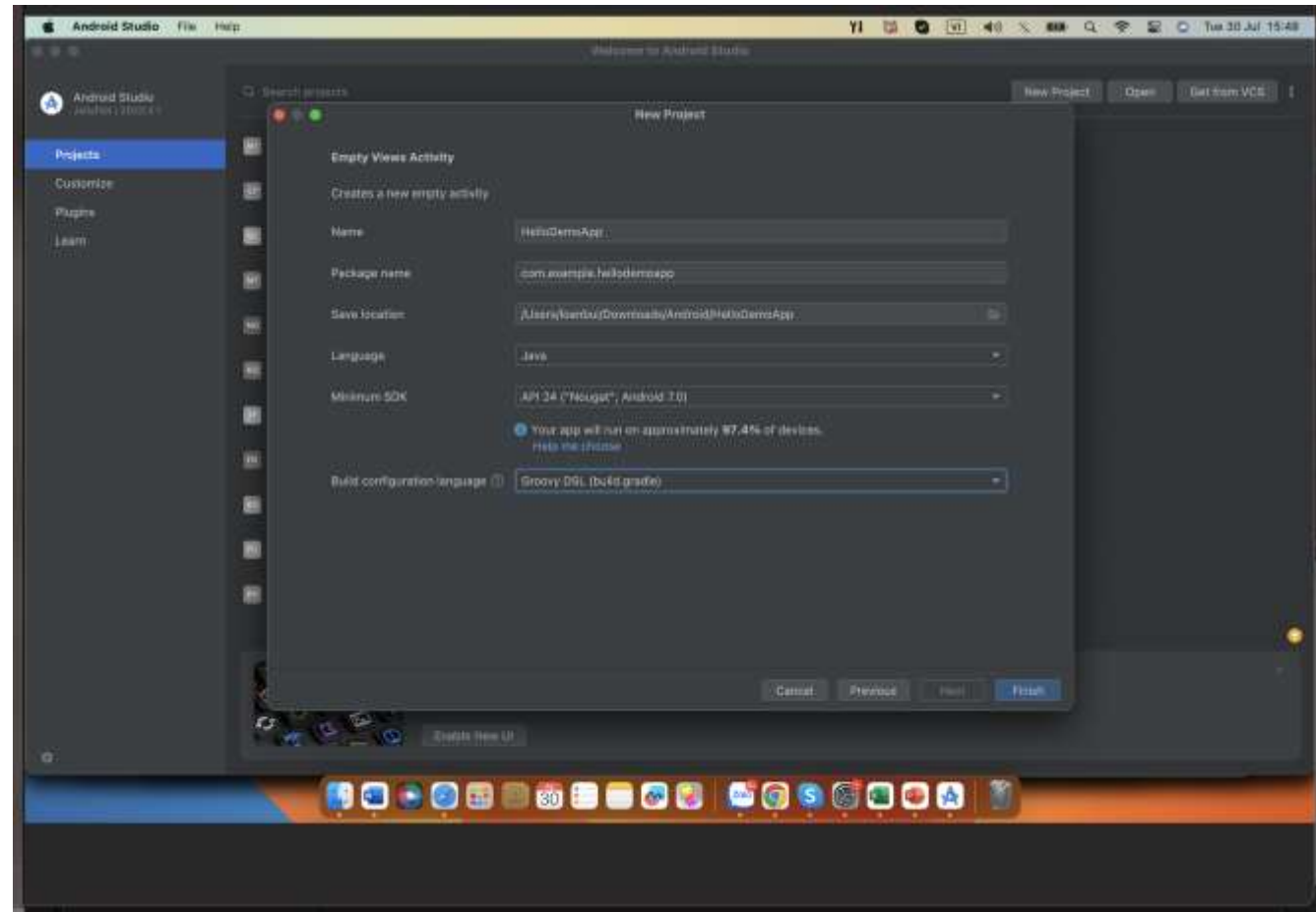
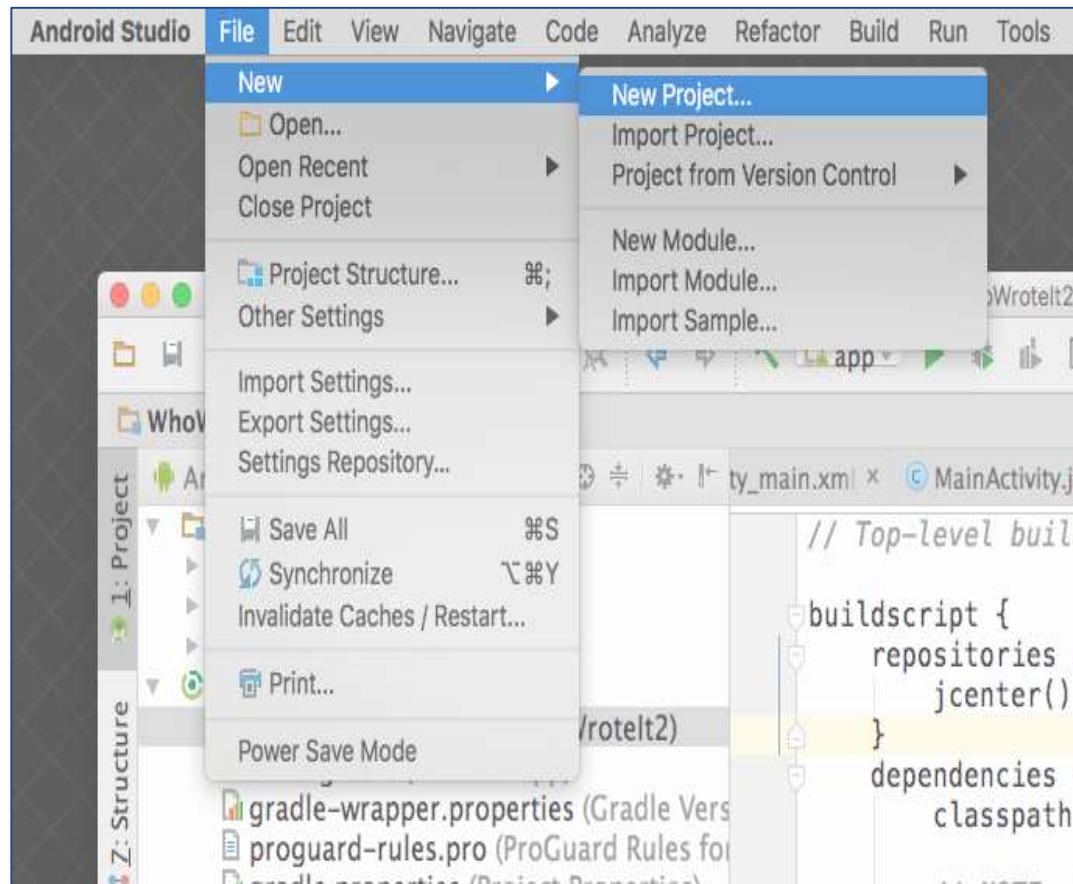
This file defines how ProGuard optimizes and obfuscates your code

Build the first application

Start Android Studio

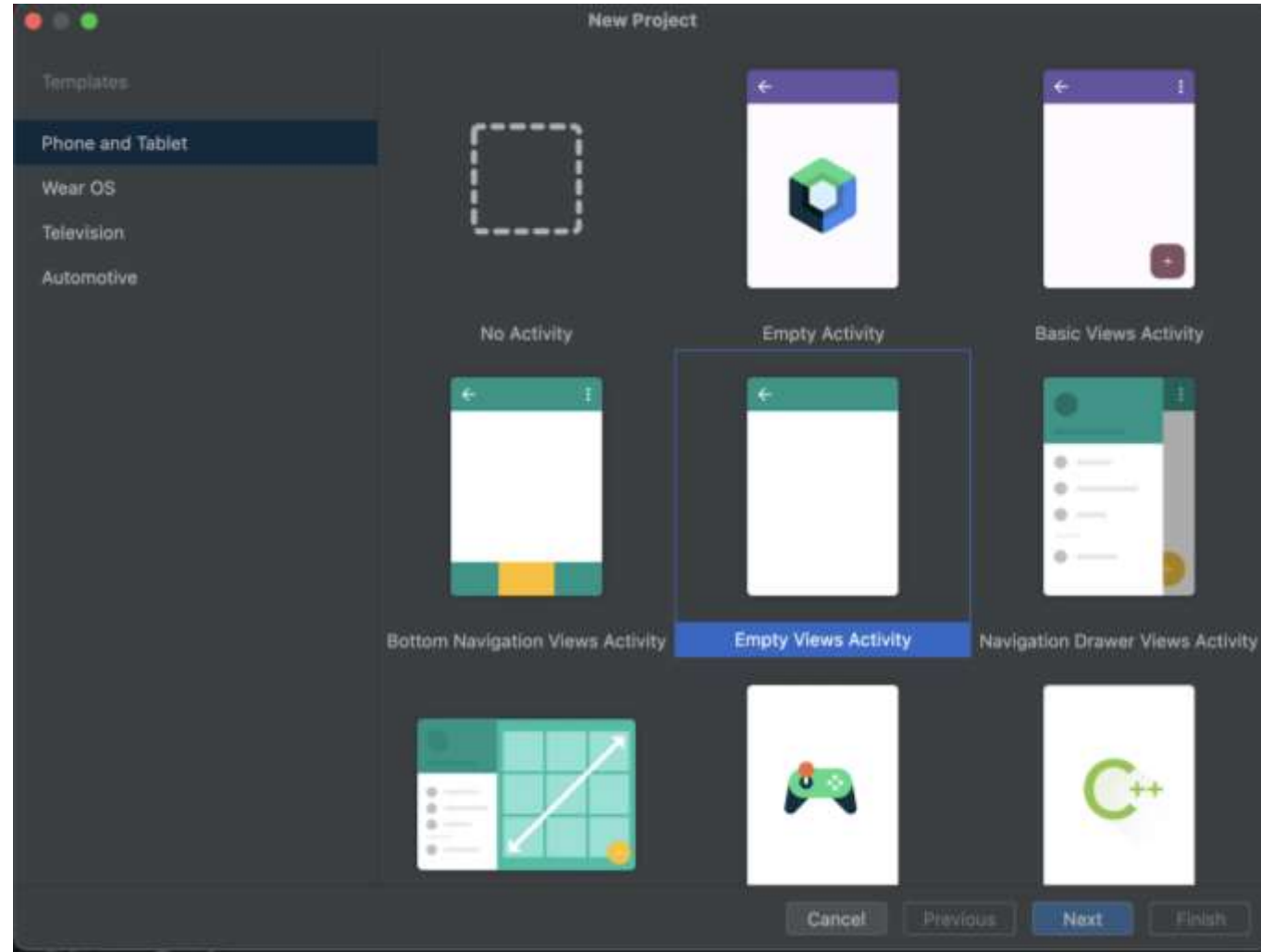


Create a project inside Android Studio



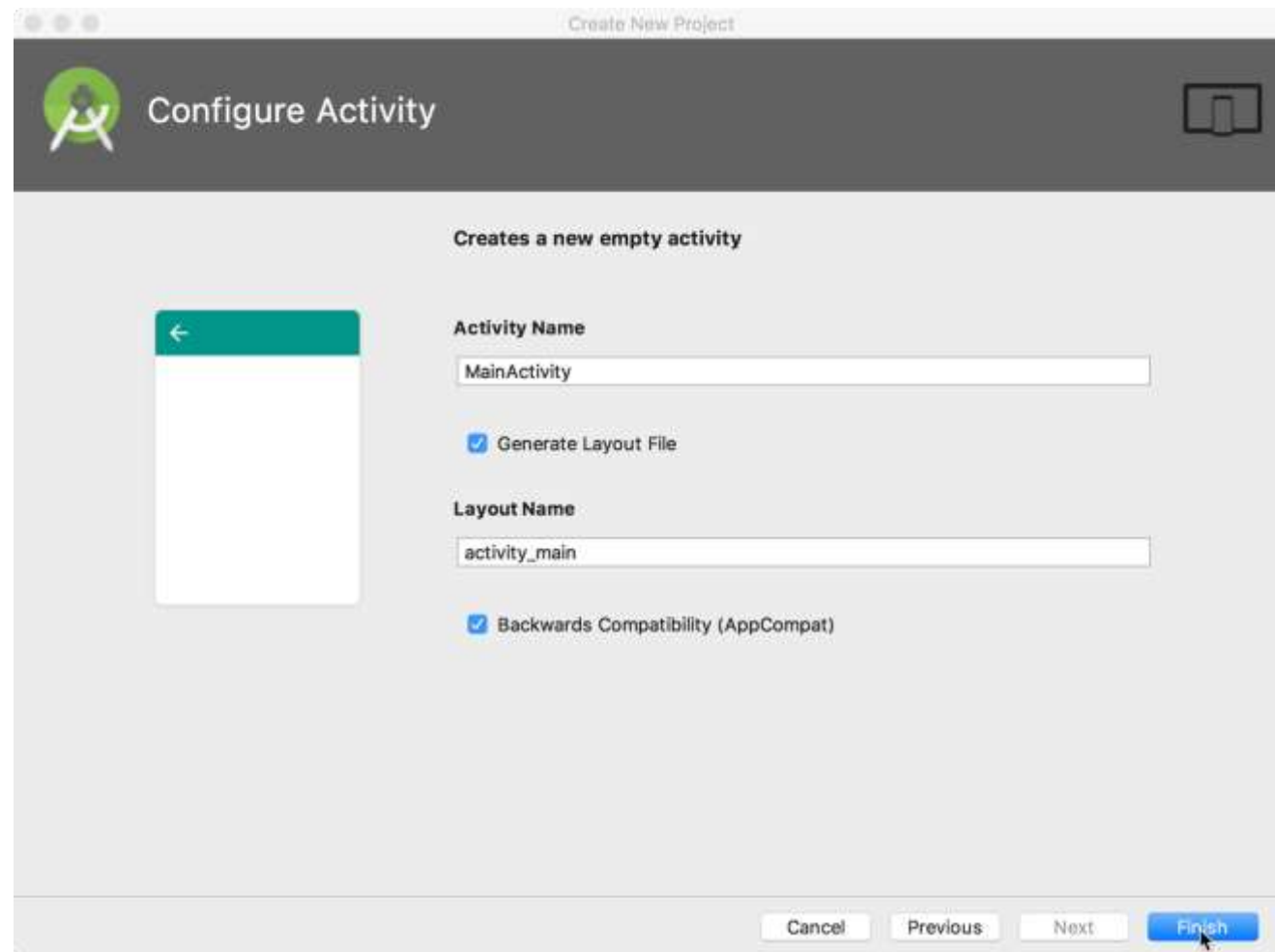
Pick activity template

- Choose templates for common activities, such as maps or navigation drawers.
- Pick Empty Activity or Basic Activity for simple and custom activities.



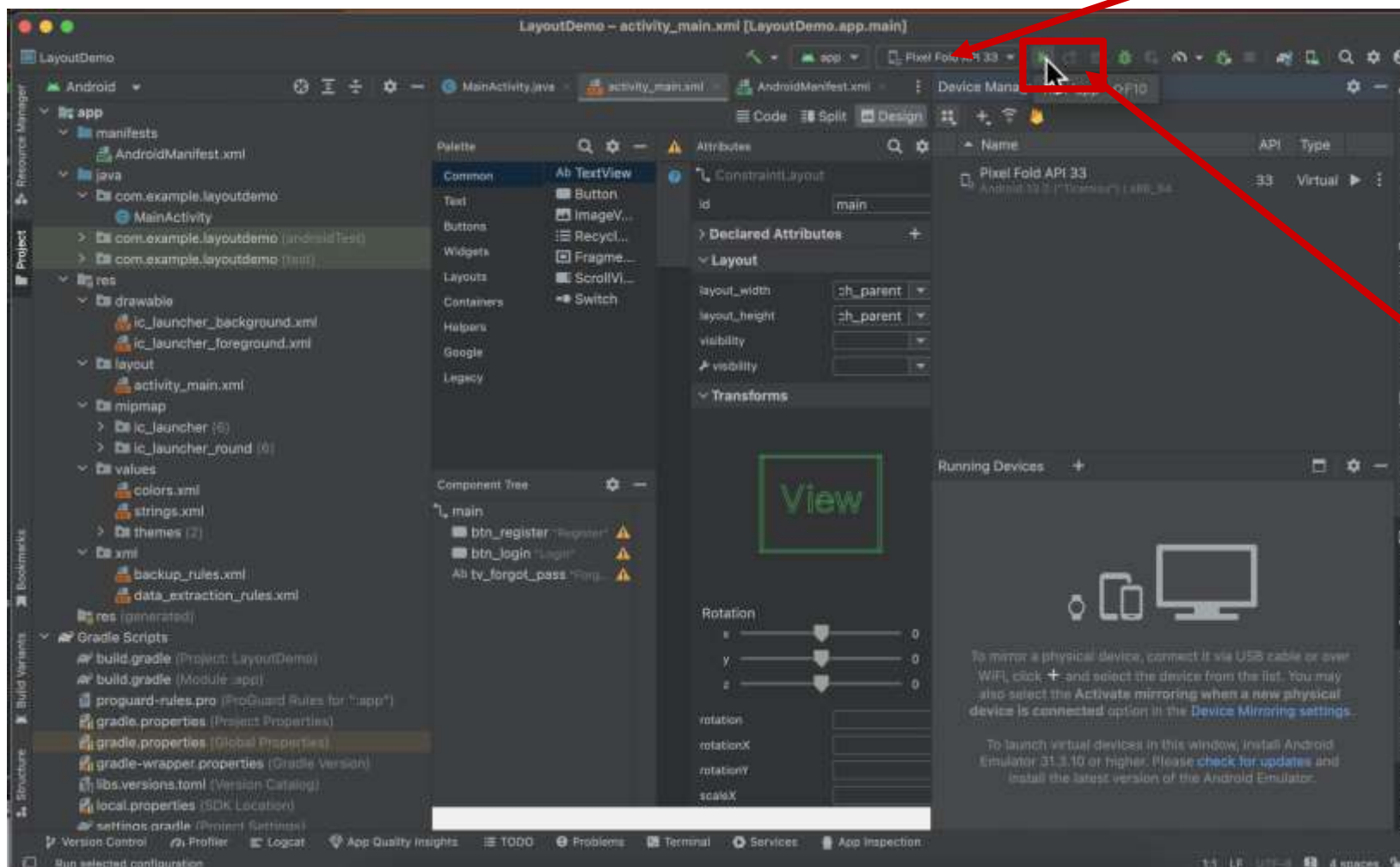
Name your activity

- Good practice:
 - Name main activity `MainActivity`
 - Name layout `activity_main`
- Use AppCompatActivity
- Generating layout file is convenient



Run your app

1. Select virtual or physical device

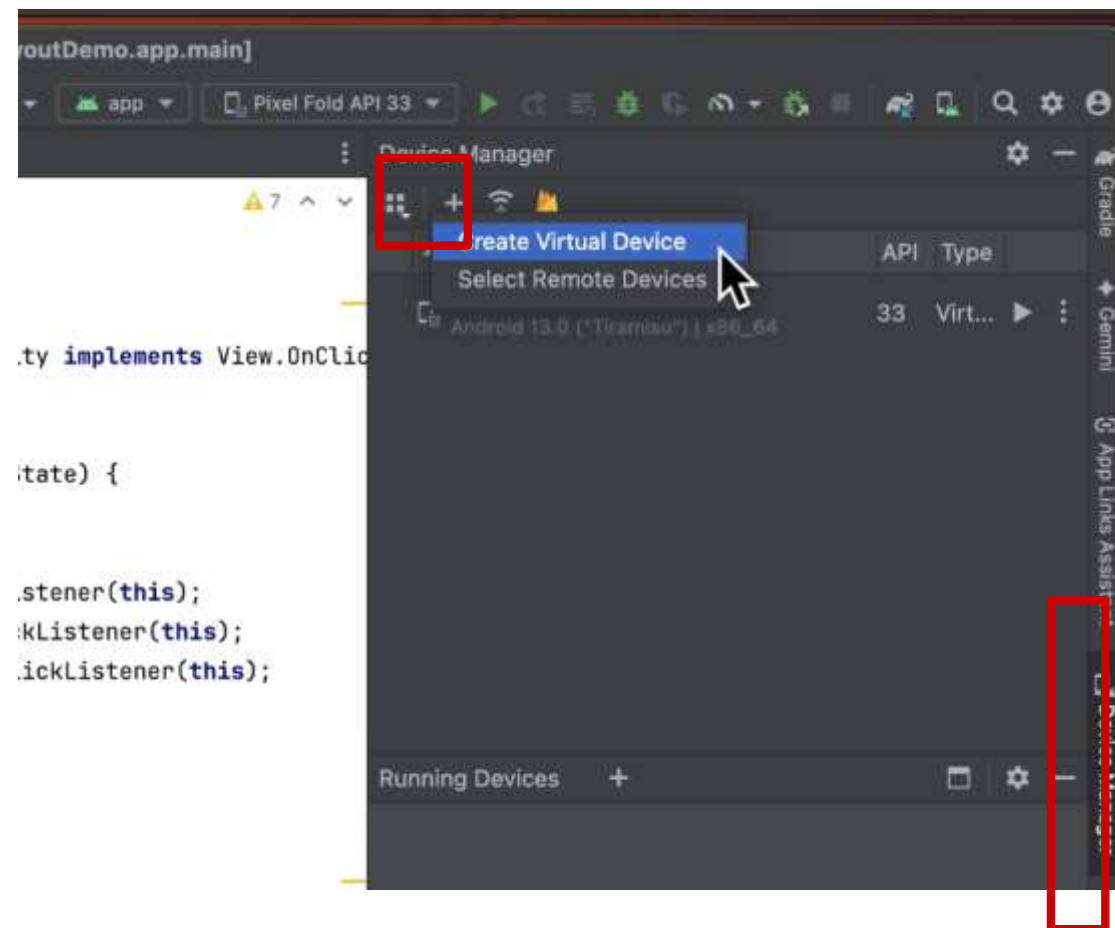


2. Run

Create a virtual device

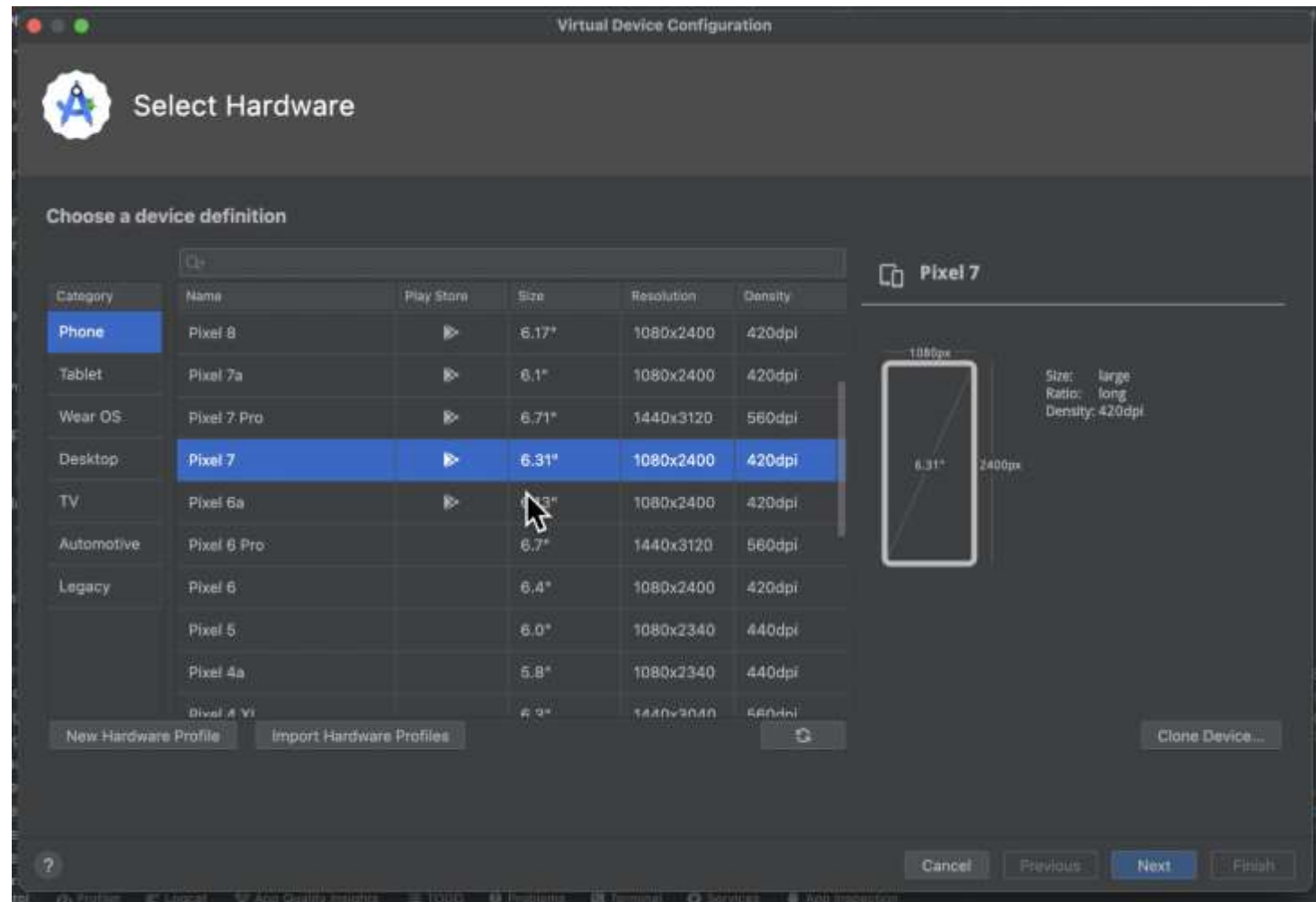
Use emulators to test app on different versions of Android and form factors.

Device Manager > Create Virtual Device

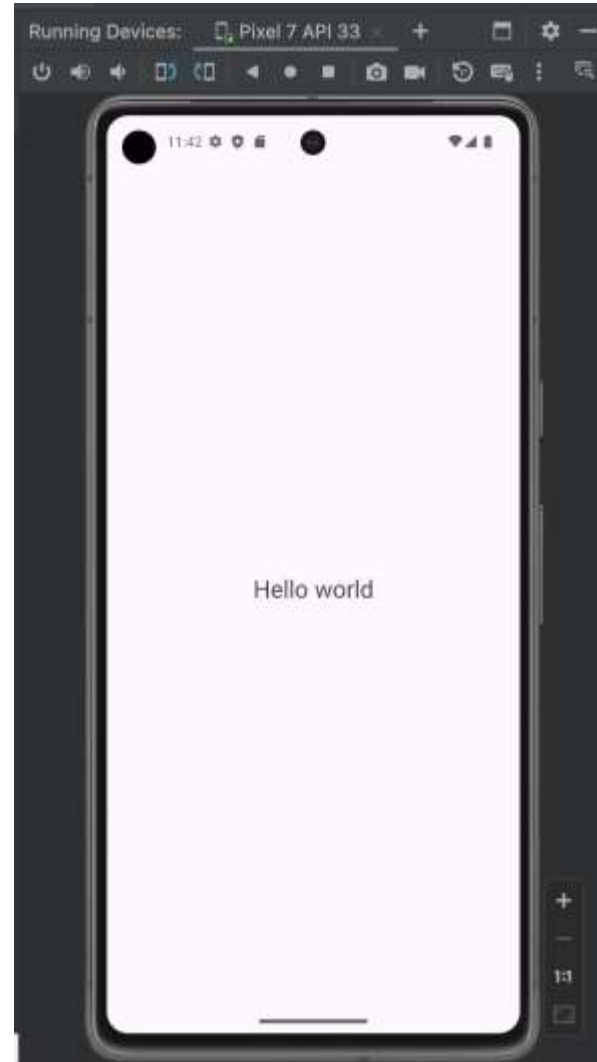
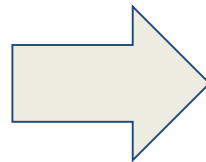


Configure virtual device

1. Choose hardware
2. Select Android version
3. Finalize

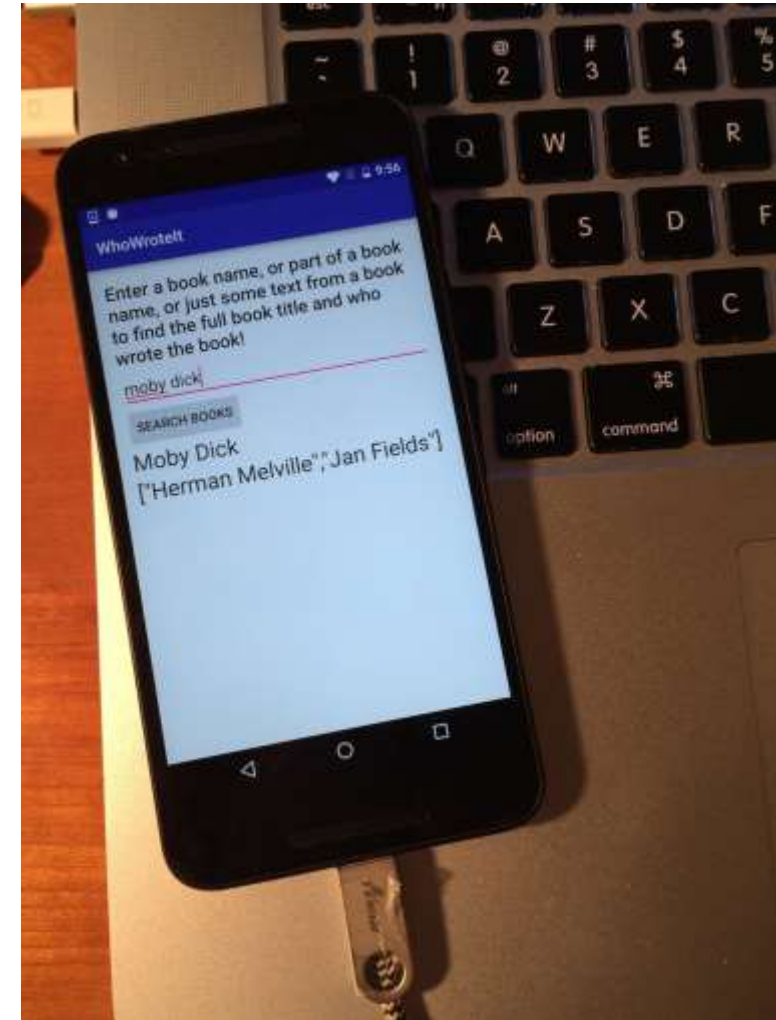


Run on a virtual device



Run on a physical device

1. Turn on Developer Options:
 - a. Settings > About phone
 - b. Tap Build number seven times
2. Turn on USB Debugging
 - a. Settings > Developer Options > USB Debugging
3. Connect phone to computer with cable



Android Application Components

- **Activities** – visual user interface focused on a single thing a user can do
- **Services** – no visual interface – they run in the background
- **Broadcast Receivers** – receive and react to broadcast announcements
- **Content Providers** – allow data exchange between applications

Activities

- Basic component of most applications
- Most applications have several activities that start each other as needed
- Each is implemented as a subclass of the base *Activity* class

Activities – The View

- Each activity has a default window to draw in (although it may prompt for dialogs or notifications)
- The content of the window is a view or a group of views (derived from View or View Group)
- Example of views: buttons, text fields, scroll bars, menu items, check boxes, etc.
- View(Group) made visible via `Activity setContentView()` method.

Services

- Does not have a visual interface
- Runs in the background indefinitely
- Examples
 - Network Downloads
 - Playing Music
 - TCP/UDP Server
- You can bind to an existing service and control its operation

Broadcast Receivers

- Receive and react to broadcast announcements
- Extend the BroadcastReceiver class
- Examples of broadcasts:
 - Low battery, power connected, shutdown, time zone changed, etc.
 - Other applications can initiate broadcasts

Content Providers

- Makes some of the application data available to other applications
- It's the only way to transfer data between applications in Android (no shared files, shared memory, pipes, etc.)
- Extends the `ContentProvider` class
- Other applications use a `ContentResolver` object to access the data provided via a `ContentProvider` object

Classwork

- Build and Run a Hello app

References

- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>